

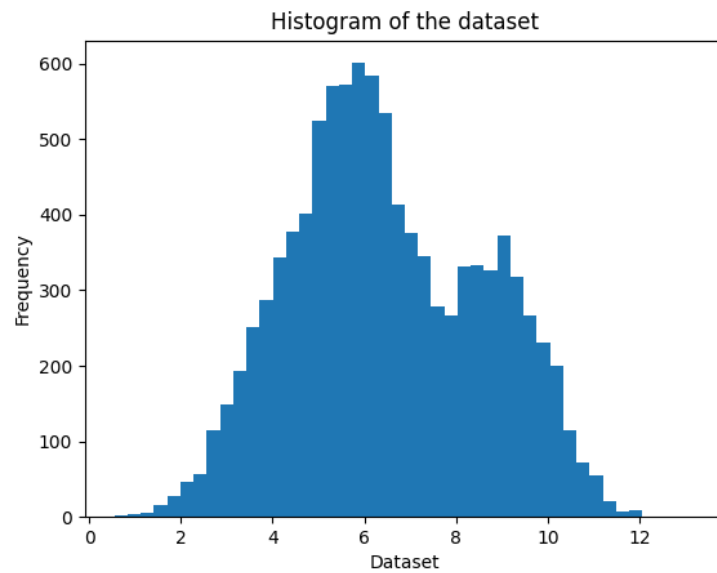
CS215 Assignment-1

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1. Fitting Data Task B

Solution: Graphically the the normal distribution parameter μ seems to be around 6.



Histogram plot

Task C

Solution: First moment of Binomial Distribution, μ_1^{bin}

$$\mu_1^{bin} = E(X) = \sum_{x=0}^n x \binom{n}{x} p^x q^{n-x}$$

$$\mu_1^{bin} = np \sum_{x=1}^n \binom{n-1}{x-1} p^{x-1} q^{n-x}$$

$$\mu_1^{bin} = np(q + p)^{n-1}$$

$$\mu_1^{bin} = np$$

Second moment of Binomial Distribution, μ_2^{bin}

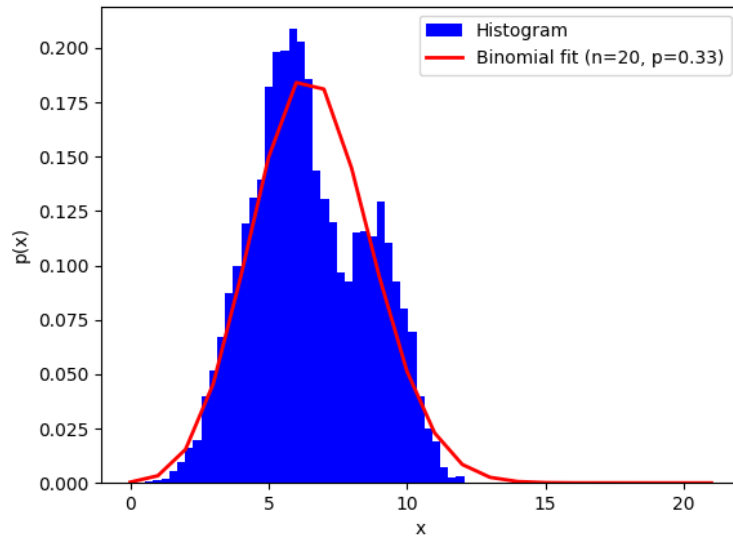
$$\mu_2^{bin} = E(X^2) = \sum_{x=0}^n x^2 \binom{n}{x} p^x q^{n-x}$$

$$\mu_2^{bin} = \sum_{x=0}^n \{x(x-1) + x\} \binom{n}{x} \binom{n-1}{x-1} \binom{n-2}{x-2} p^x q^{n-x}$$

$$\mu_2^{bin} = n(n-1)p^2 \left\{ \sum_{x=2}^n \binom{n-2}{x-2} p^{x-2} q^{n-x} \right\} + np$$

$$\mu_2^{bin} = n(n-1)p^2(q+p)^{n-2} + np$$

$$\mu_2^{bin} = n(n-1)p^2 + np$$



Binomial Fitting

Task D

Solution: Lets calculate i^{th} moment of gamma function given by:

$$f(x, \alpha, \theta) = \frac{x^{\alpha-1}}{\Gamma(\alpha)\theta^\alpha} e^{-\frac{x}{\theta}}$$

$$E[X^i] = \int_0^\infty \frac{x^k}{\Gamma(\alpha)\theta^\alpha} x^{\alpha-1} e^{-\frac{x}{\theta}} dx = \frac{1}{\Gamma(\alpha)\theta^\alpha} \int_0^\infty x^{(\alpha+i)-1} e^{-\frac{x}{\theta}} dx$$

$$\begin{aligned}
&= \frac{\Gamma(\alpha + i)\theta^{\alpha+i}}{\Gamma(\alpha)\theta^\alpha} \cdot \int_0^\infty \frac{1}{\Gamma(\alpha + i)\theta^{\alpha+i}} x^{\alpha+i-1} e^{-\frac{x}{\theta}} dx \\
&= \frac{\Gamma(\alpha + i)\theta^{\alpha+i}}{\Gamma(\alpha)\theta^\alpha} \cdot \quad (\text{p.d.f. of } \Gamma(\alpha + i, \theta) \text{ integrates to 1}) \\
&= \frac{\Gamma(\alpha + i)\theta^{\alpha+i}}{\Gamma(\alpha)\theta^\alpha} = \frac{\theta^i \Gamma(\alpha + i)}{\Gamma(\alpha)} \\
&= \frac{\theta^i (\alpha + i - 1)(\alpha + i - 2) \cdots \alpha \Gamma(\alpha)}{\Gamma(\alpha)} = \theta^i (\alpha + i - 1) \cdots \alpha.
\end{aligned}$$

Thus,

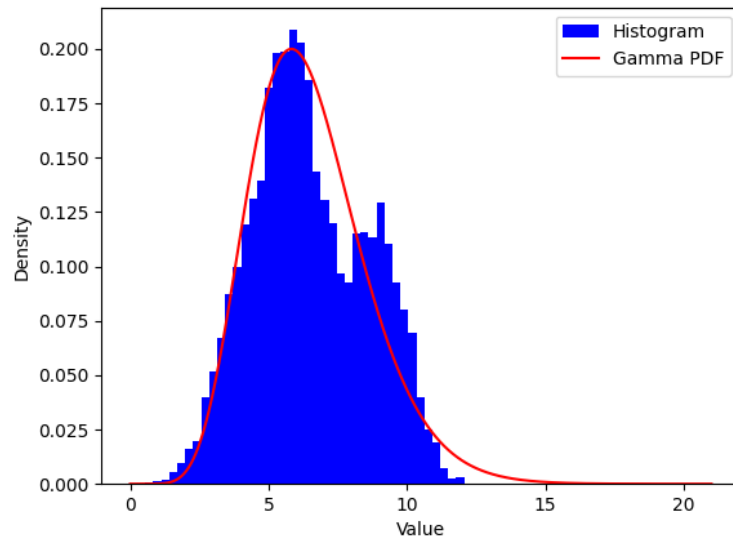
First moment μ_1^{gamma} is given by (replacing α by k in above derivation),

$$\mu_1^{gamma} = k\theta$$

And second moment μ_2^{gamma} is given by,

$$\mu_2^{gamma} = k(k + 1)\theta^2$$

Gamma Distribution (k=9.691205541246163, theta=0.6703134703624323)



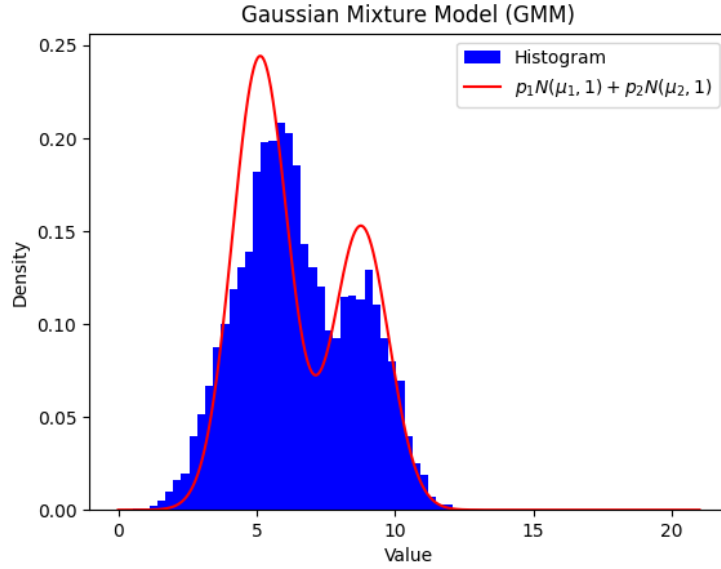
Gamma Fitting

Task E

Solution: The likelihood of binomial (-2.1570681154346776) is a bit greater (closer to zero) than likelihood of gamma distribution (-2.1608217722067953) indicating that binomial is a better fit.

Task F

Solution: The likelihood of Mix Guassian distribution obtained is -2.183038744911319. This however is not better than any of the previous two!!



GMM Fitting

Teaser (Ungraded Bonus)

Solution: In order to get a better fit, let's choose again a mixture of gaussian distribution. The difference from previous case is that instead of two gaussian, let's take three distribution superimposed over each other with probabilities p_1, p_2 and p_3 , mean as μ_1, μ_2 and μ_3 and variance of all of them as 1. The two new moments (fifth and sixth) can be derived giving the following expression:

$$\begin{aligned} \text{moment}5^{th} &= \mu^5 + 10\mu^3\sigma^2 + 15\mu\sigma^4 \\ \text{moment}6^{th} &= \mu^6 + 15\mu^4\sigma^2 + 45\mu^2\sigma^4 + 15\sigma^6 \end{aligned}$$

Similar to the previous method, calculating 5^{th} and 6^{th} moment of given dataset and solving the equations using `fsolve` function of `scipy`, we were getting impossible solutions as the jacobian failed to converge.

Thus making an initial guess and hypertuning the probabilities of the GMM, we finally arrived at a better fit whose GMM is as follows:

$$Y \sim p_1 \cdot N_1(\mu_1, 1) + p_2 \cdot N_2(\mu_2, 1) + p_3 \cdot N_3(\mu_3, 1)$$

$$\text{where, } p_1 = 0.5018740341610936$$

$$\mu1 = 6.02960769428528$$

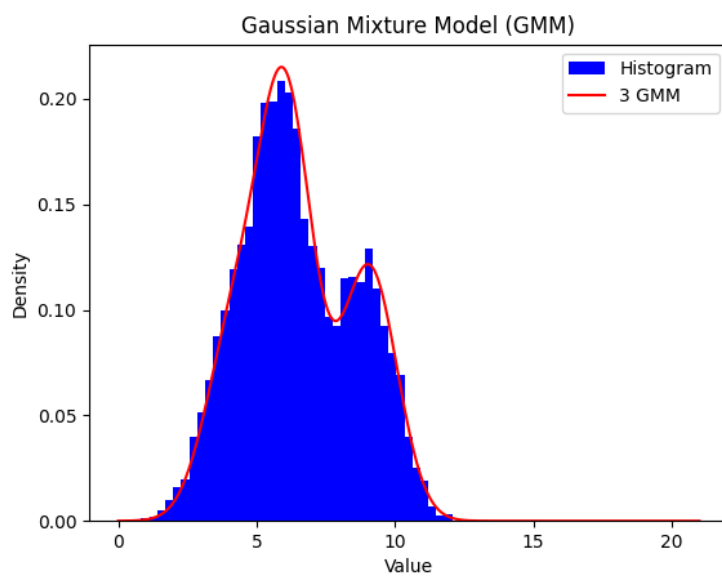
$$p2 = 0.2996456511927248$$

$$\mu2 = 9.07436305441652$$

$$p3 = 0.200116472923$$

$$\mu3 = 4.1000265473$$

Thus the final fit is:



Final GMM Fitting